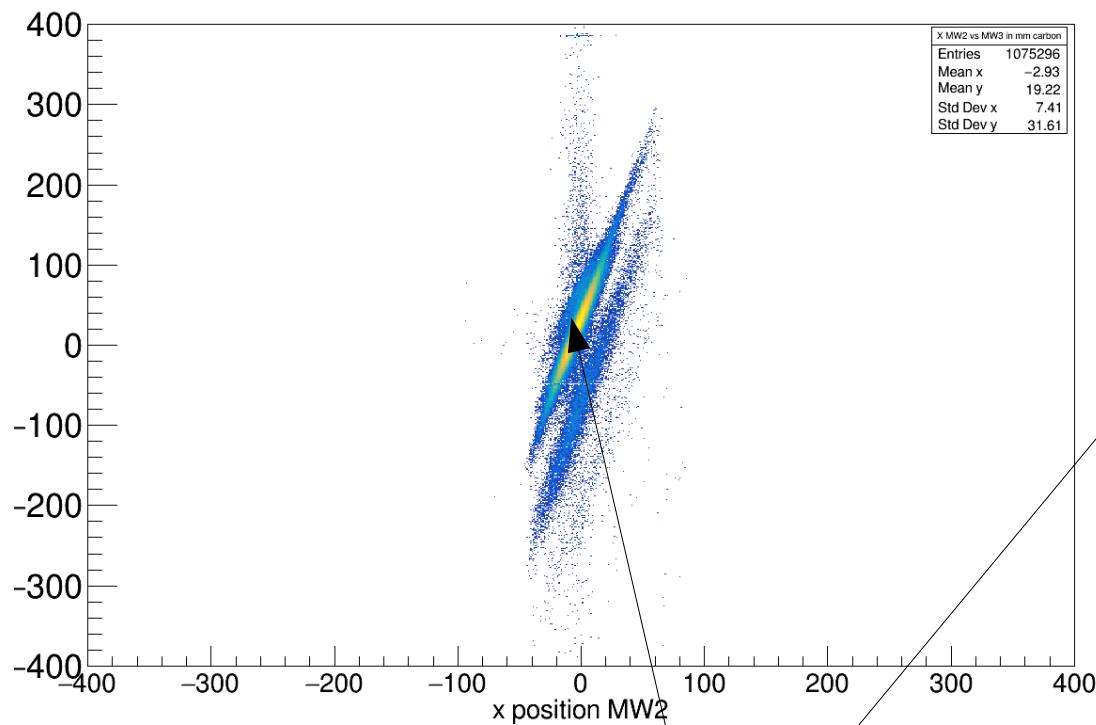
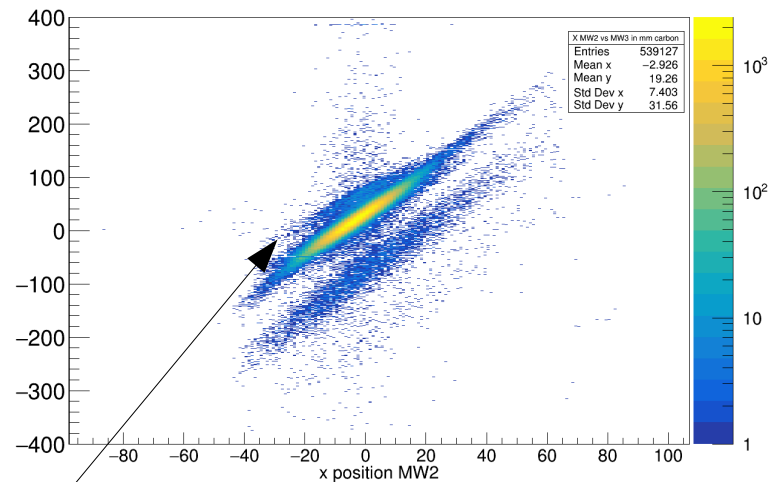


S444 Isotope Correction - Fits

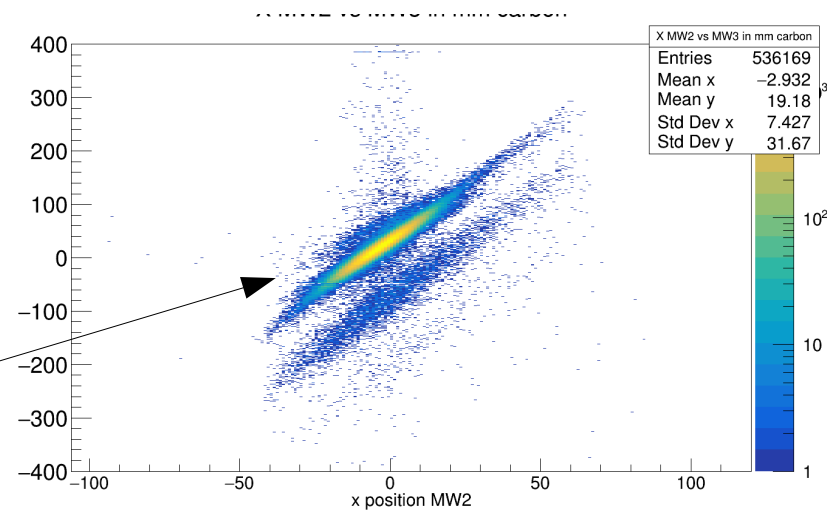
Were settings changed during Run ?



1st half



2nd half



NO, same strange line..

Bug found in code for histogram rotation...

```
rot_histo-> SetBinContent(rot_binX,rot_binY,vec_content[binX-1][binY-1]);
```

→ **this code overwrites bin content if there is already something in bin!**

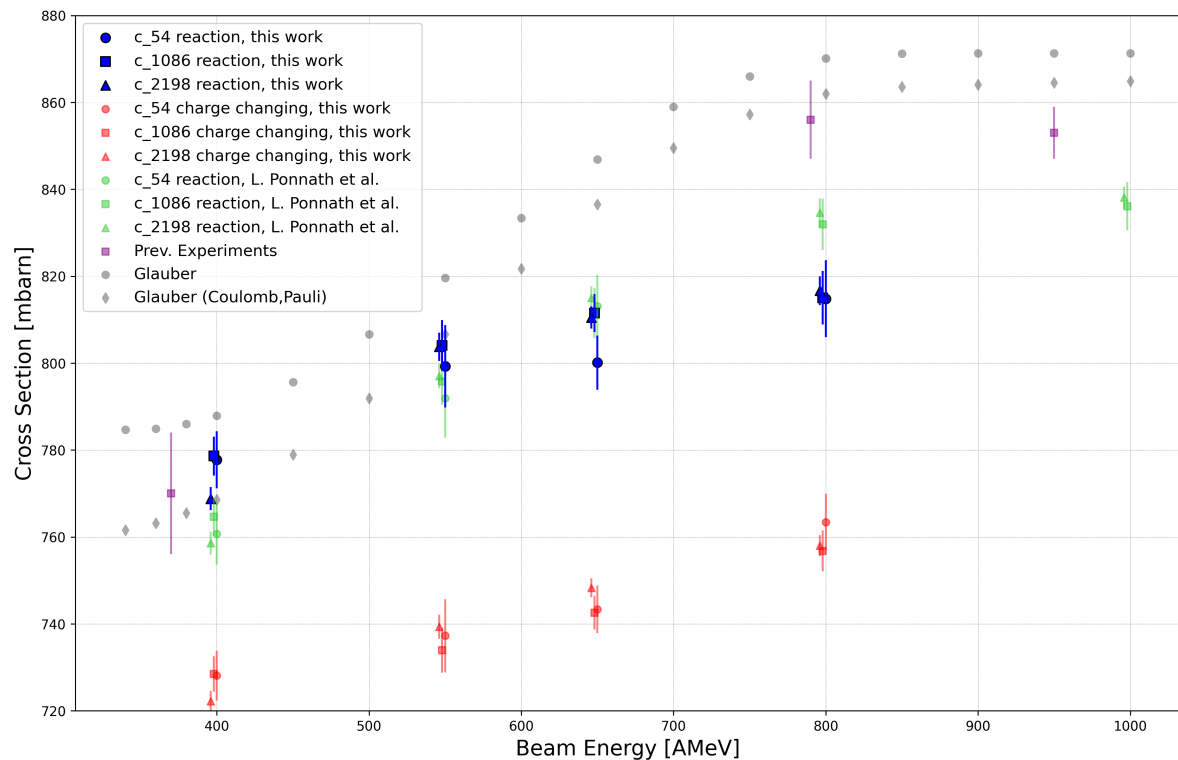
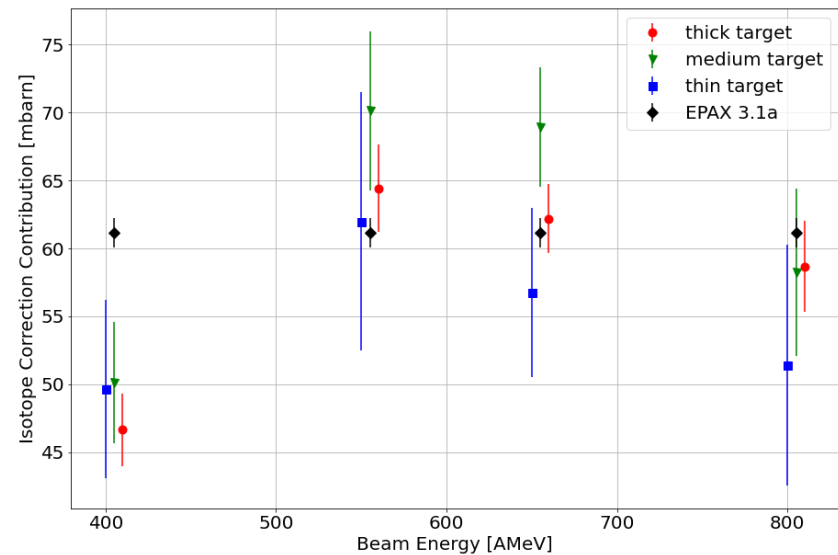
Fix:

```
double prev = rot_histo->GetBinContent(rot_binX, rot_binY);
```

```
rot_histo->SetBinContent(rot_binX, rot_binY, prev + vec_content[binX-1][binY-1]);
```

Method1: voigt fitting revised

Isotope correction:



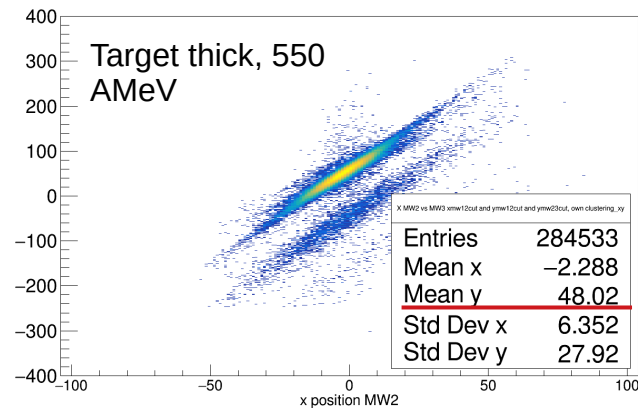
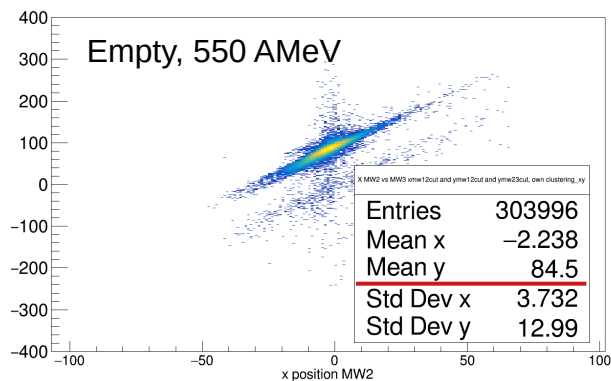
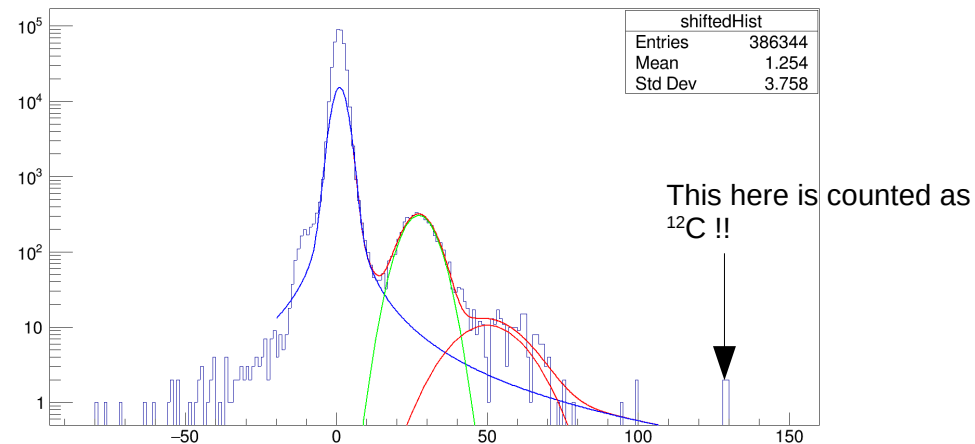
Drawbacks of method:

Correction factor (for both empty and target run) is calculated by:

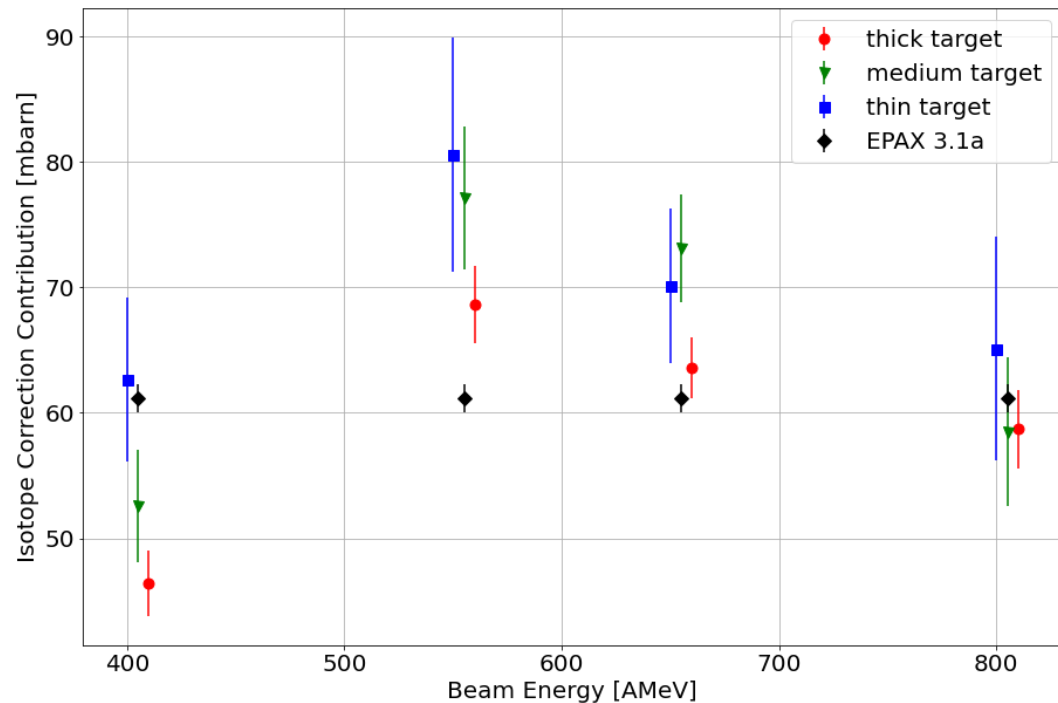
$$1 - (N_{^{11}\text{C}} + N_{^{10}\text{C}}) / N_{\text{histo}}$$

→ if I do not consider left side: no eff. Correction between target/empty!

→ however, there is anyway no real correction between target empty; histogram represents something like angular deflection, not position!



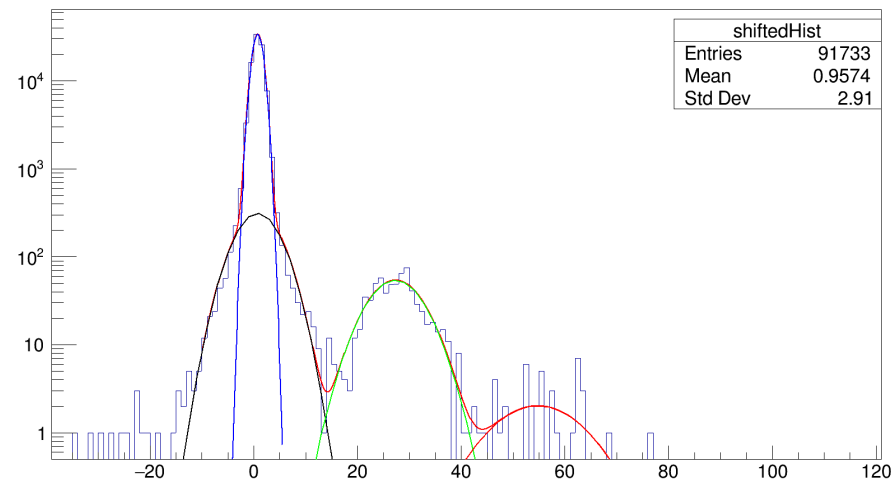
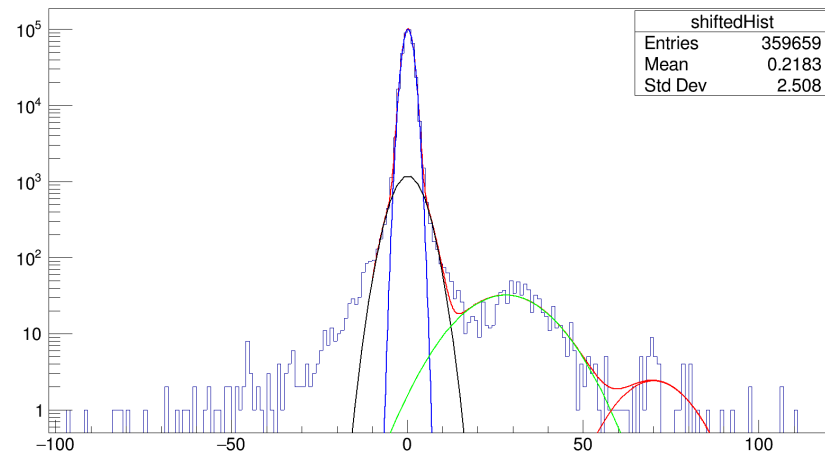
Method2: Fit with gaussians only



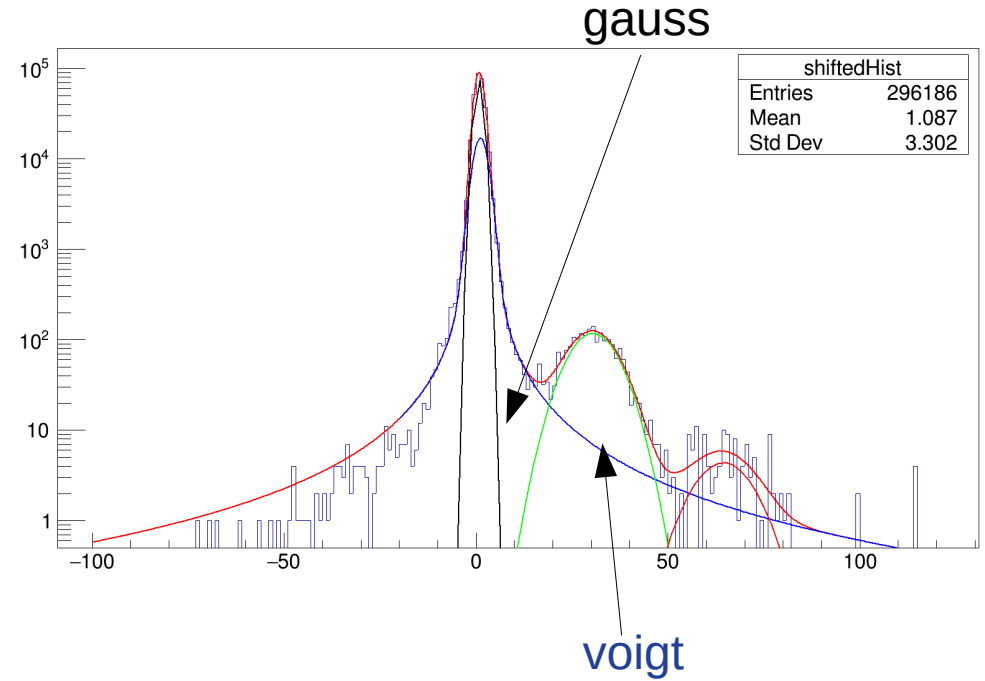
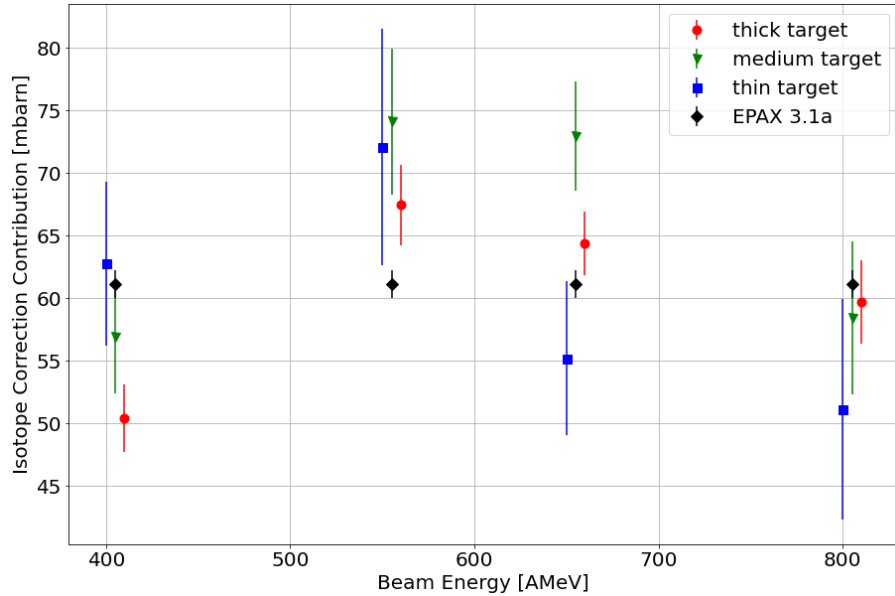
Correction factor (for both empty and target run) is calculated by:

$$1 - (N_{\text{11C}} + N_{\text{10C}}) / N_{\text{full_fit}}$$

Gaussians do not well fit the data..., e.g.:



Method3: Fit over full range with voigt +gauss

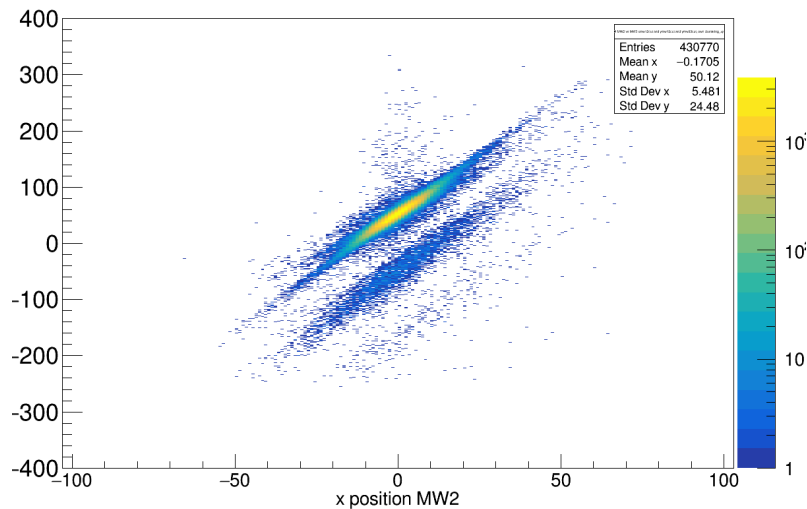


Correction factor (for both empty and target run) is calculated by:

$$1 - (N_{^{11}\text{C}} + N_{^{10}\text{C}}) / N_{\text{full_fit}}$$

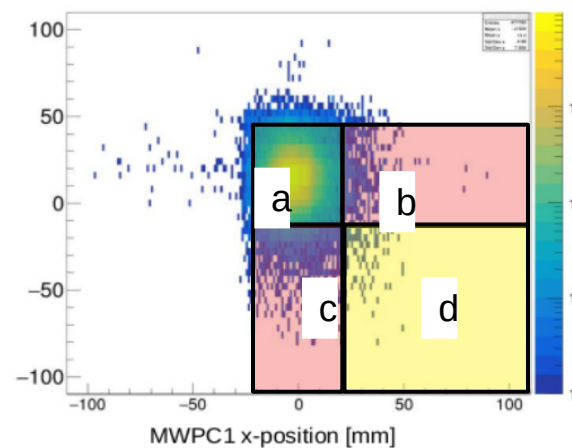
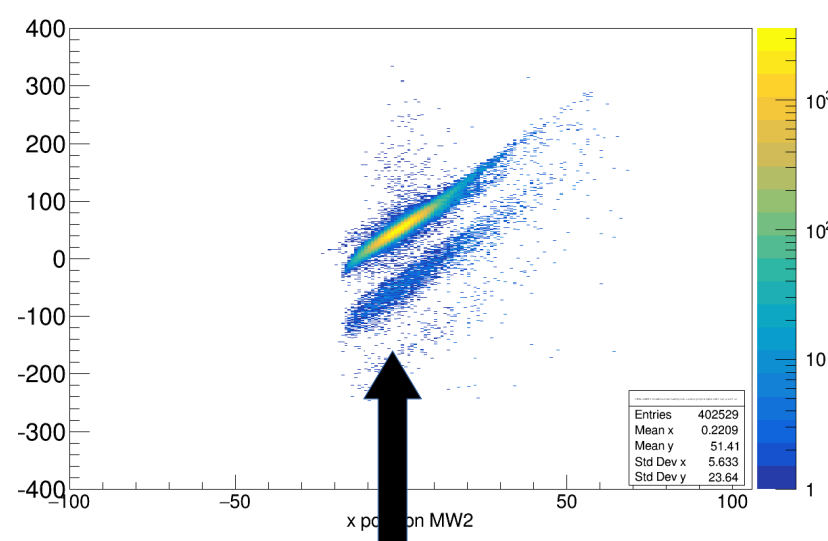
More sources of error:

I do actually not rotate and project:



This can create asymmetries in the rotated and projected plots too...

But:



Weights:

- a) 1x
- b) 2x
- c) 2x
- d) 4x